

Predictive Modeling Plan for Credit Delinquency (Geldium)

Step 1: Model Logic

Chosen Model: Logistic Regression is used to predict delinquency risk as it provides interpretable probability outputs and works well for binary classification problems.

Top 5 Features: Credit Utilization Ratio, Missed Payments, Annual Income, Debt-to-Income Ratio, Credit History Length.

Pipeline: Data Input → Feature Selection → Data Preprocessing → Model Training → Prediction → Output (Risk Score).

Alternative Models: Random Forest or XGBoost can be used for higher accuracy but are less interpretable.

Step 2: Model Justification

Logistic Regression is suitable because it balances accuracy and interpretability. In financial services, transparency is critical for regulatory compliance and decision-making. It is easy to deploy, fast, and allows understanding of feature impact, unlike complex models.

Step 3: Evaluation Strategy

Metrics: Accuracy, Precision, Recall, F1 Score, AUC-ROC.

Fairness Checks: Evaluate performance across income and demographic groups to avoid bias.

Bias Mitigation: Remove sensitive features, reweight data, and conduct audits.

Interpretation: Improve model if recall is low (missing defaulters) or precision is low (false alarms).

Final Summary

The model predicts delinquency risk using key financial indicators. Logistic Regression is chosen for interpretability and compliance. Evaluation ensures accuracy and fairness for reliable predictions.